

Your grade: 100%

Your least: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Let θ be the unknown parameter representing the number of customers that visit a store on the day of a sale. The number of customers that make a purchase is $Y \mid \theta \sim \text{Binomial}(n, \theta)$, where θ is the known probability of making a purchase, given that the customer visited the store. 2 / 2 points

True or False: The posterior distribution in this scenario will be a continuous distribution

- ☐ True
☒ False

Nice work
Nice!

2. Which of the following best describes the role of Monte Carlo simulation in Bayesian inference? 2 / 2 points
- ☐ It is not relevant for Bayesian inference.
☒ It is used to approximate the posterior predictive distribution when direct calculation is difficult.
☐ It is only used in frequentist methods like bootstrapping.
☐ It replaces the likelihood function in Bayesian analysis.

Nice work
Nice!

3. A financial institution is monitoring online transactions for fraud. Historical data suggests that only **0.1% of transactions are fraudulent**. A machine-learning model flags a transaction as suspicious if it believes the probability of fraud is high. The model has: 3 / 3 points
- **True positive rate (sensitivity):** 95% (detects 95% of actual fraud cases)
 - **False positive rate:** 5% (incorrectly flags 5% of non-fraudulent transactions)

If a transaction is flagged as suspicious, what is the posterior probability that it is actually fraudulent?

- ☒ Approximately **2%**, since fraud is rare and false positives dominate.
☐ **50%**, since the model is equally likely to make errors in either direction.
☐ Approximately **95%**, since the model is highly sensitive.
☐ **0.1%**, since that is the prior probability of fraud.

Nice work

4. A hospital is studying the survival rate of patients given a new treatment for a rare disease. Based on historical data, the **prior distribution** for survival probability (μ) is **Beta(5,5)**, representing initial uncertainty (a prior mean of 50%). 3 / 3 points

After treating **10 patients, 7 survived**. What is the **posterior predictive probability** that the next patient will survive?

- ☐ **0.5** (the mean of the prior)
☒ **0.6**
☐ **0.7** (the estimate from observed data)
☐ **0.67**, since Bayesian inference ignores prior beliefs when data is available.

Nice work

Correct! The posterior is a Beta(12,8) and the mean is 12/(12+8) = 0.6

5. A pollster wants to estimate the proportion of voters favoring a candidate. The initial belief is that the support is around 50% but uncertain. A poll of 1,000 people finds 540 in favor. The frequentist MLE estimate of support is $\frac{540}{1000} = 0.54$. How might a Bayesian estimate differ? 3 / 3 points
- ☐ Bayesian methods are not useful for estimating proportions.
☐ The Bayesian estimate would be higher than 0.54 regardless of the prior.
☐ The Bayesian estimate would always be exactly 0.54, just like the MLE.
☒ If the prior was strongly centered at 50%, the Bayesian estimate would likely be pulled closer to 0.50.

Nice work
Nice!

6. A meteorologist wants to predict the probability of rain tomorrow using Bayesian inference. They have: 3 / 3 points
- A prior belief that rain occurs **30%** of the time in similar conditions.
 - A model based on atmospheric pressure that suggests a **60%** chance of rain.

They use Monte Carlo simulation to generate 10,000 samples from the posterior predictive distribution. What is the primary reason for using this approach?

- ☐ The Monte Carlo method is necessary because Bayesian inference cannot be performed algebraically.
☐ Monte Carlo methods always give the exact probability of an event.
☒ It allows incorporating both prior knowledge and new data uncertainty to estimate rain probability.
☐ The Monte Carlo method ensures that the posterior predictive distribution is always normally distributed.

Nice work
Nice work!

7. **Select all that apply:** The posterior predictive distribution... 3 / 3 points
- ☐ can be estimated by randomly generating values from the prior distribution, and then using those values as parameters in the posterior distribution.
☒ can be estimated by randomly generating values from the posterior distribution, and then using those values as parameters in the sampling distribution/likelihood.

Nice work

- ☐ has less variability than the posterior distribution.
☒ is the distribution over new observations given previous observations.

Nice work

- ☒ accounts for uncertainty in the parameters when making a prediction.

Nice work

8. For a six word dictionary, this problem assumes we are dealing with a language where: 3 / 3 points
- $W := \{\text{huz, suu, slt, sat, fan, for}\}$

An extensive study of the literature written in this language reveals the following:

- All words are equally likely except that the word "for" is θ times as likely as any other word.
- Each keystroke is an error with probability θ .
- All letters are equally likely to produce errors.
- Given that a letter is typed incorrectly, it is equally likely to be replaced by any other letter.
- Errors are independent across letters.

For example:

- The probability of correctly typing "fun" (or any other word) is $(1 - \theta)^3$.
- The probability of typing "sun" or "fon" when intending to type "fun" is $\theta(1 - \theta)^2$.
- The probability of typing "foo" or "non" when intending to type "fun" is $\theta^2(1 - \theta)$.

Let $I(\text{word})$ denote that the "word" was intended. Suppose that we want to compute the probability that the intended word was "fun" given that "sun" was actually typed, i.e., $P(I(\text{huz}) \mid \text{suu})$, using Bayes' theorem. Which probabilities among those below are needed?

(A) The prior probability of intending to type "sun".

(B) The probability of typing "sun" given that we intended to type "fun".

(C) The probability of intending to type "fun" given that we actually typed "sun".

- ☐ (A) only.
☒ (B) only.
☐ (A) and (B) only.
☐ All of the above.

Nice work

Some prior probabilities are needed.

9. Consider the following modeling scenario: For an entire year, researchers collect data on fraudulent credit card transactions, including whether or not a particular transaction was ruled as fraudulent, the amount of each purchase, the distance from the card holder's zip code, whether the purchase was online or not, and several other variables. The goal is to use this data to construct a model that will help flag future purchases as potentially fraudulent. 3 / 3 points

Based on the information given, a reasonable first attempt at a model would be:

- ☐ A binomial regression, with the distance from the card holder's zip code as the response and all other variables as predictors.
☐ A binomial regression, with the online/not online variable as the response and all other variables as predictors.
☐ A binomial regression, with the amount of each purchase as the response and all other variables as predictors.
☒ A binomial regression, with the fraudulent/not fraudulent variable as the response and all other variables as predictors.

Nice work

10. For a six word dictionary, this problem assumes we are dealing with a language where: 3 / 3 points
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- The probability of typing "foo" or "non" when intending to type "fun" is $\theta^2(1 - \theta)$.

Let $I(\text{word})$ denote that the "word" was intended.

What is the probability of typing "sun" given that the intended word was "fun"?

☐ $\theta^2(1 - \theta)$

☐ θ^3

☒ $\theta(1 - \theta)^2$

Nice work
Correct!